



FraudLens

TEAM WEBNEST

Project report · real-time scam protection for mobile money

FraudLens stops scam payments **before the money leaves the wallet**. It checks **where the money is going**, not just who is sending it, and picks the gentlest step that will stop the scam.

86.9%

scams caught

0.53%

honest payments
interrupted

96.6%

victims' money
stopped*

3.4 ms

to decide (p95)

11x

better than rules

1 The problem

The victim presses send

- A caller pretends to be **upay staff**, a lottery office or an investment agent.
- The victim pays from **their own phone, with their own PIN**.
- The money goes to a **mule wallet** and is cashed out at an agent in minutes.

Why today's checks miss it

- They only check the **sender**, and the sender looks normal.
- Hand-written rules are right only **1 time in 8**.
- Blocking after the fact is too late. The money is already gone.

2 Our solution

One question, answered in milliseconds: is this payment going to a scammer, and if so, what is the gentlest thing that will stop it?

Send

Customer
presses send



Features

61 live signals



Models

payment · mule
wallet · anomaly



Policy

thresholds + 7
rules



Tier

allow · warn ·
step-up · hold



Analyst

verdict becomes
a training label

3 Four gentle tiers

Tier	What happens	Who decides	False alerts / day
Allow	Goes through. Nothing shown. 98.5% of payments.	nobody	–
Warn	Scam warning in Bangla. Customer may continue.	the customer	27.8
Step-up	Verify again, then a 30-minute cooling-off.	the customer	8.4
Hold	Paused, with a 30-minute review deadline.	a human analyst	2.3

Out of about 5,400 scored payments a day. Each row counts that tier and stricter.

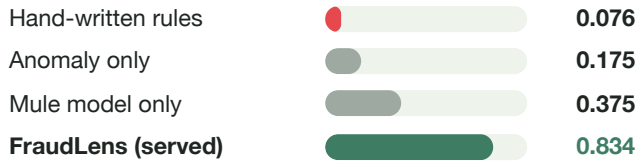
টাকা পাঠানোর আগে একটু থামুন। প্রতারকরা উপায় কর্মকর্তা, লটারি কর্তৃপক্ষ বা বিনিয়োগ এজেন্ট সেজে এ ধরনের ওয়ালেটে টাকা পাঠাতে বলে। আপনি কি এই ব্যক্তিকে চেনেন?

“Pause before you send. Scammers pose as upay staff, lottery officials or investment agents... Do you know this person?”

* Counting holds on the mule's cash-out. All results come from a 25-day test period of synthetic data (134,545 payments, ৳35.5 lakh at risk).

4 Results

Ranking quality vs. rules (PR-AUC)



Rules are right 1 in 8 times. FraudLens is right 8 in 10 at the same alert budget.

Scams caught at the warn tier, by type

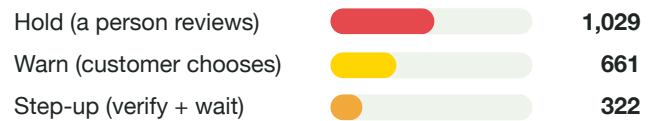


The investment scam was hidden from training. With cash-out holds, **92.3%** of its money is still stopped.

Victims' money stopped

Tier and above	At payment	+ cash-out hold
Hold	72.8%	83.1%
Step-up	78.1%	90.1%
Warn	85.1%	96.6%

Where the 2,012 interruptions land



98.5% of payments are never interrupted.

5 Beyond single payments

70

Mules caught early

94 of 145 mule wallets found, 70 **before any victim paid**, a median **40.2 h** ahead.

34

Scam rings

34 rings, 331 wallets linked by shared phones and transfers. 94% are true fraud wallets.

90.9%

Risky agents

Precision with **no labels**. All commission-farming agents found.

0.77 → 0.91

Learns from analysts

Analyst verdicts become labels. Retraining raised PR-AUC on later data.

98.5%

Shadow mode

A new model scores every payment silently and agreed on the tier 98.5% of the time.

90.2%

Scam messages

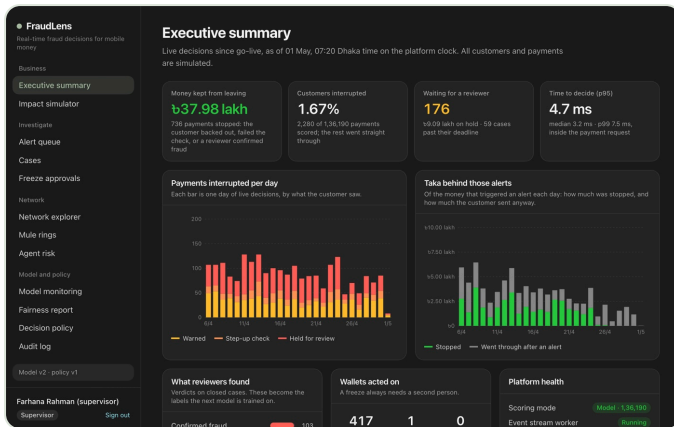
Flags scam SMS in English, Bangla and Banglish, with 0.23% false flags.

6 Explained, not a black box

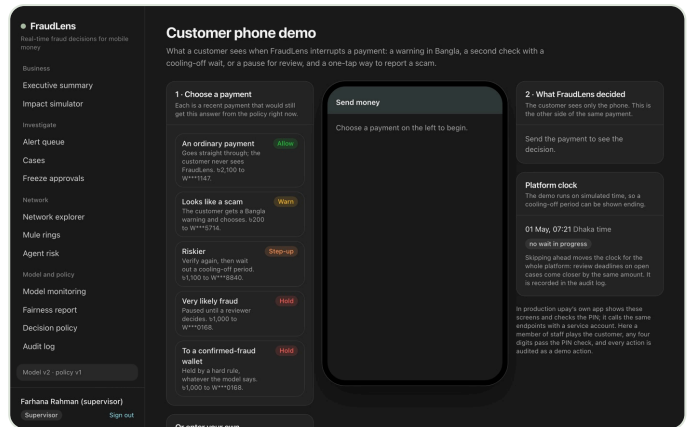
- **Ranked reasons** (SHAP) as plain sentences, in **Bangla and English**
- The exact **rule trace** and the **closest past case**

- One case per mule wallet, not one alert per victim
- Optional AI wording is rejected if it adds any number or ID not in the evidence

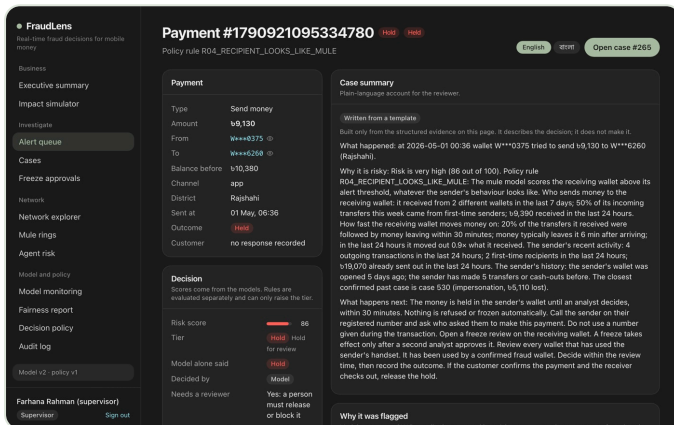
7 The console we built



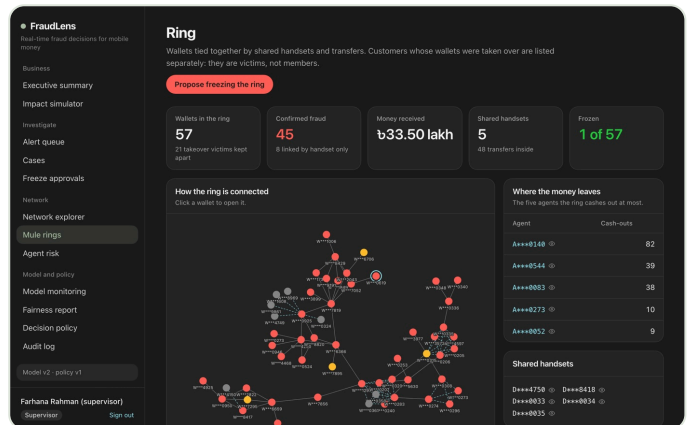
Executive summary. Money stopped, customers interrupted, backlog, latency.



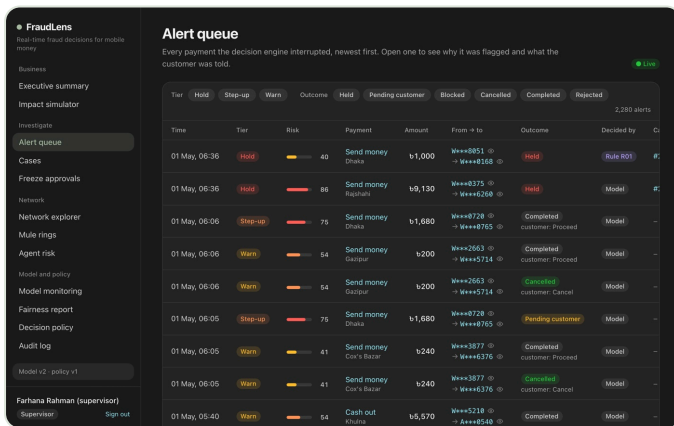
Customer phone demo. The real scoring API, next to FraudLens's decision.



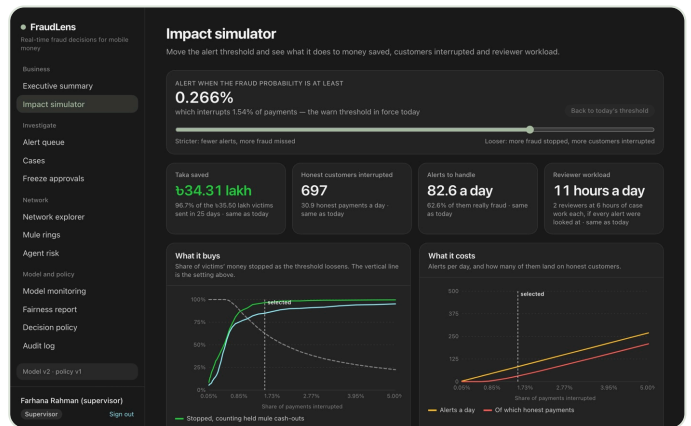
Why it was flagged. Reasons, rule trace and case summary in English or বাংলা.



Mule rings. Wallets tied by shared phones and transfers.



Alert queue. Live feed. Wallet numbers stay masked until an audited reveal.



Impact simulator. Slide the threshold and watch the money saved and the people interrupted.

8 Tech stack

Layer	Tools
Models	Python 3.12, LightGBM, scikit-learn, SHAP
API	FastAPI, SQLAlchemy, Alembic, PostgreSQL, Redis Streams, server-sent events
Console	Next.js 16, React 19, Tailwind CSS v4, d3-force
Ops	Docker Compose, Make, uv, pnpm, Playwright smoke tests, GitHub Actions CI

9 Architecture

Offline · make pipeline

- Simulator: 20,000 customers, 600 agents, 120 days
- 5 scam types, 1 held out of training
- 61 features, LightGBM, Isolation Forest
- Thresholds from alert budgets

Online · make api

- POST /v1/score for live answers
- Redis Streams for events
- Same features, models and policy as training
- PostgreSQL: decisions, cases, audit log

Console + MLOps

- Next.js console, no business logic
- Verdicts and two-person freezes
- Retrain, shadow mode, drift
- A person promotes every model

10 Trust: checked, not promised

Served = evaluated

Payments compared	136,180
Tier differences	0
Feature differences	0
Failed events	0
Backend tests	227

Guardrails in code

- The model **never blocks money** alone
- A freeze needs **two people** (API + DB check)
- Audit log **cannot be edited**
- Wallet numbers **masked** by default
- Rules-only fallback if a model is down

Fairness, stated plainly: wallets under 30 days old get more false alarms (6× when sending, 16× when receiving). The fairness report tracks this.

11 Impact

Customers

A warning in their own language at the one moment it can still help.

Fraud teams

One case per mule wallet, with the reasons and network ready.

upay

A clear trade-off between money saved, people interrupted and review hours.

12 Limits and next steps

Honest limits

- All data is **synthetic**. Numbers will change on real traffic.
- Prototype: one scorer process, no TLS or scaling yet.
- Bangla texts are not yet professionally reviewed.

Next

- A quiet **shadow test on real upay payments**
- Re-fit thresholds on real data
- Scale out, harden security, review the Bangla texts

Run it all with one command: **make demo** · *The victim presses send, so look at where the money is going.*